

📌 Module A (Lec 5–6): Steepest Descent Analysis on Quadratics

The Setup. Minimize $f(x) = \frac{1}{2}x^\top Qx - b^\top x$ where $Q \succ 0$.

• **Algorithm:** $x^{(k+1)} = x^{(k)} - \alpha_k g^{(k)}$, with $g^{(k)} = \nabla f(x^{(k)})$.

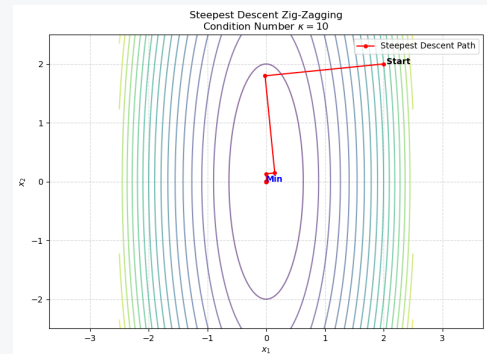
• **Exact Line Search:** $\alpha_k = \frac{g^{(k)\top} g^{(k)}}{g^{(k)\top} Q g^{(k)}}$.

Convergence Rate (Theorem 8.2). Let $V(x) = f(x) - f(x^*)$.

$$V(x^{(k+1)}) \leq \left(\frac{\lambda_{\max}(Q) - \lambda_{\min}(Q)}{\lambda_{\max}(Q)} \right) V(x^{(k)}).$$

• **Condition Number** $\kappa(Q) = \lambda_{\max}/\lambda_{\min}$ dictates speed.

• $\kappa \gg 1 \implies$ Slow, **Orthogonal (Why?) Zig-zagging** $g^{(k+1)} \perp g^{(k)}$.



🔧 Practice: Exercise 8.8 (Fixed-Step Gradient Method)

Largest stepsize range for global convergence Consider the function

$$f(x) = 3(x_1^2 + x_2^2) + 4x_1x_2 + 5x_1 + 6x_2 + 7, \quad x = [x_1, x_2]^\top \in \mathbb{R}^2.$$

Suppose that we use a fixed-step-size gradient algorithm to find the minimizer of f :

$$x^{(k+1)} = x^{(k)} - \alpha \nabla f(x^{(k)}).$$

Find the largest range of values of α for which the algorithm is globally convergent.

(a) **Quadratic structure.** Compute $\nabla f(x)$ and the Hessian $H = \nabla^2 f(x)$.

(b) **Eigenvalues & Positive Definiteness.** Compute the eigenvalues of H . Is H positive definite?

(c) **Error Iteration.** Prove $e^{(k)} := x^{(k)} - x^* \implies e^{(k+1)} = (I - \alpha H)e^{(k)}$.

(d) **Global convergence condition.** Using $e^{(k)} := x^{(k)} - x^*$ and $e^{(k+1)} = (I - \alpha H)e^{(k)}$, find the largest range of α such that $\rho(I - \alpha H) < 1$.

Module B (Week 3 & Chapter 9): Newton's Method in $\mathbb{R}^n$

The Idea. Locally approximate f with a **quadratic** Taylor expansion and jump to its minimum.

$$x^{(k+1)} = x^{(k)} - \frac{f'(x^{(k)})}{f''(x^{(k)})} \implies x^{(k+1)} = x^{(k)} - [\nabla^2 f(x^{(k)})]^{-1} \nabla f(x^{(k)}).$$

Let $H_k = \nabla^2 f(x^{(k)})$ and $g_k = \nabla f(x^{(k)})$. The step solves the linear system $H_k d_k = -g_k$.

Properties.

- **Quadratic convergence** ($p = 2$) near x^* .
- Finds the minimum of a **perfect quadratic** in **1 step**.
- **Pitfall:** If $H_k \not\prec 0$, the step might not be a descent direction (attracts to saddle points).

Practice: Exercise 9.1 (Newton's Method)

$f(x) = (x - x_0)^4$: **convergence and order:** Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = (x - x_0)^4$, where $x_0 \in \mathbb{R}$ is a constant. Suppose that we apply Newton's method to the problem of minimizing f , with iterates $x^{(0)}, x^{(1)}, x^{(2)}, \dots$

(a) Write down the update equation for Newton's method applied to the problem.

(b) Let $y^{(k)} = |x^{(k)} - x_0|$. Show that the sequence $\{y^{(k)}\}$ satisfies $y^{(k+1)} = \frac{2}{3}y^{(k)}$.

(c) Show that $x^{(k)} \rightarrow x_0$ for any initial guess $x^{(0)}$.

(d) Show that the order of convergence of the sequence $\{x^{(k)}\}$ in part (b) is 1.

(e) Theorem 9.1 states that under certain conditions, the order of convergence of Newton's method is at least 2. Why does that theorem not hold in this particular problem?

Pitfalls

- **Gradient Descent:** "Steepest" $\neq$ "Straight to minimum" (unless $\kappa = 1$). It often zig-zags.
- **Newton's Method:** Expensive for large n (requires H^{-1}).
- **Global vs Local:** Both methods find *local* minimizers (or stationary points).